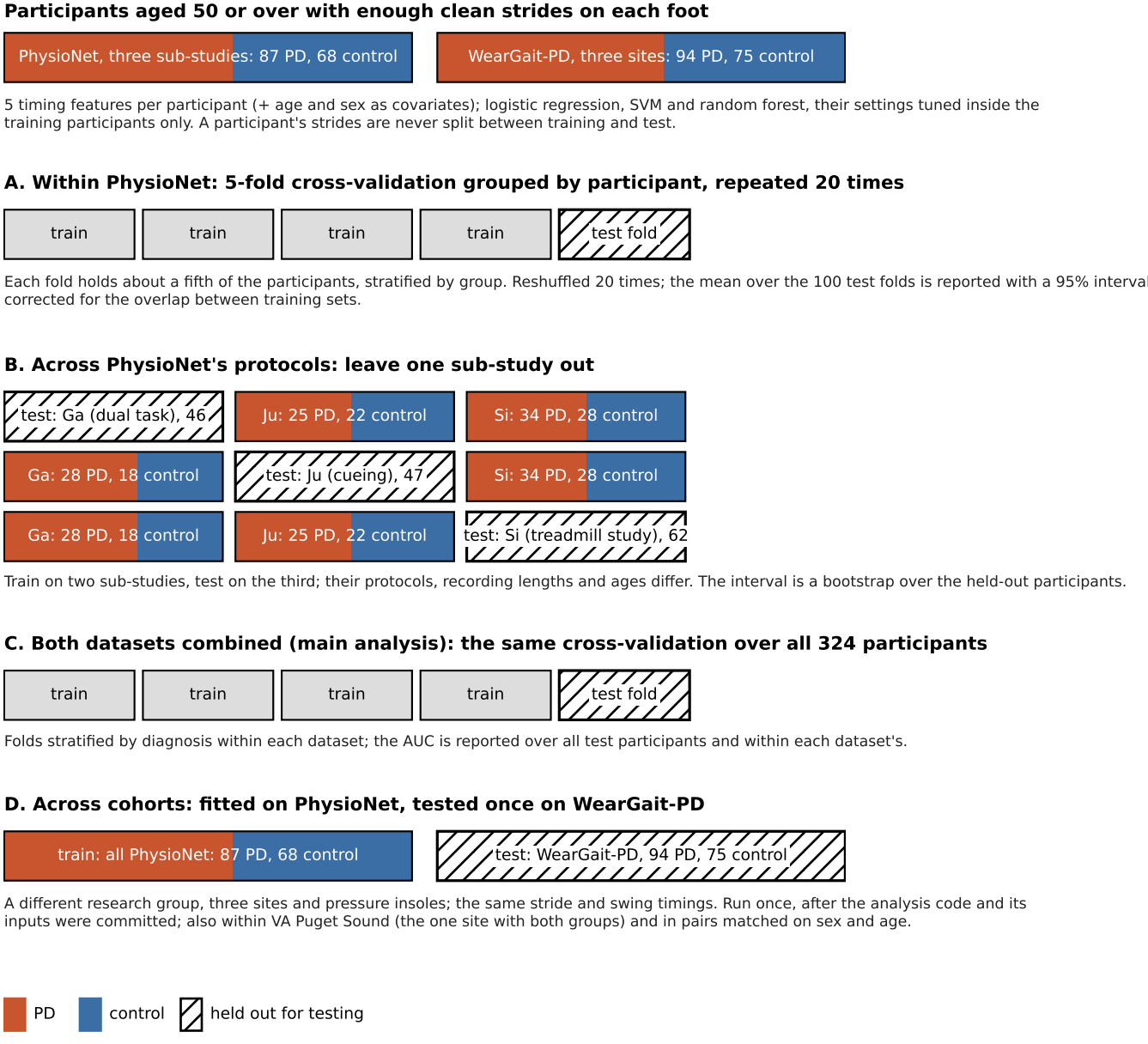
Participants aged 50 or over with enough clean strides on each foot
PhysioNet, three sub-studies: 87 PD, 68 control
WearGait-PD, three sites: 94 PD, 75 control
5 timing features per participant (+ age and sex as covariates); logistic regression, SVM and random forest, their settings tuned inside the training participants only. A participant's strides are never split between training and test.
A. Within PhysioNet: 5-fold cross-validation grouped by participant, repeated 20 times
train
train
train
train
test fold
Each fold holds about a fifth of the participants, stratified by group. Reshuffled 20 times; the mean over the 100 test folds is reported with a 95% interval corrected for the overlap between training sets.
B. Across PhysioNet's protocols: leave one sub-study out
test: Ga (dual task), 46
Ju: 25 PD, 22 control
Si: 34 PD, 28 control
Ga: 28 PD, 18 control
test: Ju (cueing), 47
Si: 34 PD, 28 control
Ga: 28 PD, 18 control
Ju: 25 PD, 22 control
test: Si (treadmill study), 62
Train on two sub-studies, test on the third; their protocols, recording lengths and ages differ. The interval is a bootstrap over the held-out participants.
C. Both datasets combined (main analysis): the same cross-validation over all 324 participants
train
train
train
train
test fold
Folds stratified by diagnosis within each dataset; the AUC is reported over all test participants and within each dataset's.
D. Across cohorts: fitted on PhysioNet, tested once on WearGait-PD
train: all PhysioNet: 87 PD, 68 control
test: WearGait-PD, 94 PD, 75 control
A different research group, three sites and pressure insoles; the same stride and swing timings. Run once, after the analysis code and its inputs were committed; also within VA Puget Sound (the one site with both groups) and in pairs matched on sex and age.
PD
control
held out for testing